An Infinite Wiener–Pitt Set for the Cantor Group

Candidate full solution packet

arXiv:1305.3324, Final Remarks item 5

11 August 2026

Status. Candidate full affirmative answer, subject to human review. The proof uses the finite-group integer-rounding theorem of Ohrysko–Sanders–Wojciechowski [2, Theorem 11]; the passage from finite quotients to the Cantor group and the norm-diagonal construction below are the new steps.

1 Source question

Ohrysko and Wojciechowski [1, Final Remarks, item 5] ask whether any infinite Wiener–Pitt set exists for the convolution measure algebra of the Cantor group. Their printed question appears on journal page 25:

> 5. The crucial element in our proof was the use of the Littlewood conjecture to estimate the number of repetitions of any specific value taken by the Fourier coefficients. In the case of the Cantor group, the Littlewood conjecture does not hold. This fact encourages us to ask whether any infinite Wiener - Pitt set exist for the convolution measure algebra on the Cantor group.

We use the notation

$$\mathbb{D} := \prod_{r\geq 1} \mathbb{Z}/2\mathbb{Z}, \qquad \Gamma := \widehat{\mathbb{D}} = \bigoplus_{r\geq 1} \mathbb{Z}/2\mathbb{Z}.$$

For $\mu \in M(\mathbb{D})$, its Fourier–Stieltjes transform is $\widehat{\mu}(\gamma) = \int_{\mathbb{D}} \overline{\gamma(x)}\, d\mu(x)$. A compact set $K \subset \mathbb{C}$ is a *Wiener–Pitt set* for $M(\mathbb{D})$ if

$$\widehat{\mu}(\Gamma) \subset K \quad \Longrightarrow \quad \sigma_{M(\mathbb{D})}(\mu) = \overline{\widehat{\mu}(\Gamma)}$$

for every $\mu \in M(\mathbb{D})$.

2 Proof intuition

The obstruction identified in the source paper is that one Fourier value can occur on an enormous set of characters. On the Cantor group this is not a reason to count the level set: the dual group is the increasing union of finite groups. We round on every finite subgroup, lift the resulting measure from the corresponding finite quotient, and take a weak-star limit. This turns every sufficiently separated Fourier level into an idempotent measure, with the same quantitative norm bound as in the finite theorem.

A second obstruction is that the definition of a Wiener–Pitt set places no bound on $\|\mu\|$. We handle all norms diagonally. At stage N an interpolation polynomial labels the first N Fourier levels by the integers $1, \ldots, N$ and sends a sufficiently tiny tail close to zero. For measures of norm at most N, rounding this one polynomial isolates the whole prefix and leaves a residual small enough to start an infinite idempotent expansion.

The expansion of μ itself need not converge in measure norm. Its square and cube do. A character of the measure algebra can select at most one of the orthogonal level idempotents. The square determines a level up to sign, and the cube fixes the sign, forcing every spectral value to be an actual Fourier value (or a limit of such values).

3 A rounding lemma for the Cantor group

For a real-valued function f with $d(f, \mathbb{Z}) := \sup_x \min_{m\in\mathbb{Z}} |f(x) - m| < 1/2$, write $f_{\mathbb{Z}}$ for its pointwise nearest-integer function. We use the following form of [2, Theorem 11]: there is a function

$$\delta : (0, 1] \times (0, 1] \times [1, \infty) \longrightarrow (0, 1/2)$$

such that, for every finite abelian group F, every real $f \in A(F)$ satisfying $\|f\|_{A(F)} \le M$ and $d(f, \mathbb{Z}) \le \delta(\varepsilon_1, \varepsilon_2, M)$ obeys

$$\|f_{\mathbb{Z}}\|_{A(F)} \le (1 + \varepsilon_1)\|f\|_{A(F)} + \varepsilon_2.$$

Lemma 1 (profinite integer rounding)**.** *Let $\theta \in M(\mathbb{D})$ have real Fourier–Stieltjes transform and $\|\theta\| \le M$, where $M \ge 1$. If*

$$d(\widehat{\theta}, \mathbb{Z}) \le \delta(\varepsilon_1, \varepsilon_2, M),$$

then there is $\theta_{\mathbb{Z}} \in M(\mathbb{D})$ such that

$$\widehat{\theta_{\mathbb{Z}}} = (\widehat{\theta})_{\mathbb{Z}}, \qquad \|\theta_{\mathbb{Z}}\| \le (1 + \varepsilon_1)\|\theta\| + \varepsilon_2.$$

No finiteness assumption is made on the support of $(\widehat{\theta})_{\mathbb{Z}}$.

Proof. Let $\Gamma_r \le \Gamma$ be the subgroup generated by the first r coordinates, let $H_r := \Gamma_r^{\perp}$, and let $\pi_r : \mathbb{D} \to \mathbb{D}/H_r$ be the quotient map. The finite measure $(\pi_r)_{\#}\theta$ has norm at most $\|\theta\|$, and its Fourier transform is $\widehat{\theta}|_{\Gamma_r}$. Applying the finite-group theorem to this restriction gives a measure λ_r on $\mathbb{D}/H_r$ whose Fourier transform is $(\widehat{\theta}|_{\Gamma_r})_{\mathbb{Z}}$ and such that

$$\|\lambda_r\| \le (1 + \varepsilon_1)\|\theta\| + \varepsilon_2.$$

Lift λ_r to a measure $\widetilde{\lambda}_r$ on $\mathbb{D}$ by replacing each atom at a coset $x + H_r$ with the same scalar multiple of normalized Haar measure on that coset. Then $\|\widetilde{\lambda}_r\| = \|\lambda_r\|$ and

$$\widehat{\widetilde{\lambda}_r}(\gamma) = \begin{cases} (\widehat{\theta}(\gamma))_{\mathbb{Z}}, & \gamma \in \Gamma_r, \\ 0, & \gamma \notin \Gamma_r. \end{cases}$$

The uniformly bounded net $(\widetilde{\lambda}_r)$ has a weak-star cluster point $\theta_{\mathbb{Z}}$ in $M(\mathbb{D}) = C(\mathbb{D})^*$. For each fixed $\gamma \in \Gamma$, the displayed Fourier coefficient is equal to $(\widehat{\theta}(\gamma))_{\mathbb{Z}}$ for all sufficiently large r. Hence the cluster point has the required transform, and weak-star lower semicontinuity gives the norm bound. $\square$

4 Construction of the compact set

We recursively construct a strictly decreasing sequence

$$1 = a_1 > a_2 > \cdots > 0$$

and put $K := \{0\} \cup \{a_n : n \geq 1\}$.

Suppose $a_1, \ldots, a_N$ have been chosen. Let q_N be the unique real polynomial of degree at most N satisfying

$$q_N(0) = 0, \qquad q_N(a_j) = j \quad (1 \leq j \leq N).$$

Write $q_N(t) = \sum_k c_{N,k} t^k$ and set

$$B_N := \max\left\{1, \sum_k |c_{N,k}| N^k\right\}, \qquad E_N := 2B_N + 1.$$

For $1 \leq j \leq N$, let $p_{j,N}$ be the interpolation polynomial on $\{0, 1, \ldots, N\}$ satisfying

$$p_{j,N}(j) = 1, \qquad p_{j,N}(k) = 0 \quad (k \in \{0, 1, \ldots, N\},\ k \neq j).$$

If $p_{j,N}(t) = \sum_k d_{j,N,k} t^k$, define

$$C_N := \max_{1 \leq j \leq N} \sum_k |d_{j,N,k}| E_N^k, \qquad T_N := N + C_N \sum_{j=1}^{N} a_j.$$

All these numbers are determined before a_{N+1} is chosen. Since $q_N(0) = 0$, we may choose $a_{N+1} > 0$ so small that

$$a_{N+1} < a_N/9, \tag{1}$$

$$\frac{a_{N+1}}{a_N} \leq \delta\left(\tfrac{1}{2}, \tfrac{1}{2}, a_N^{-3/2}\right), \tag{2}$$

$$\sup_{0 \leq t \leq a_{N+1}} |q_N(t)| \leq \delta(1, 1, B_N), \tag{3}$$

$$a_{N+1} \leq T_N^{-2}. \tag{4}$$

Every right-hand side is positive, so the recursion is possible. Condition (1) implies $a_n \to 0$; hence K is compact and infinite.

5 Main theorem

Theorem 2. *The set K constructed above is a Wiener–Pitt set for $M(\mathbb{D})$. In particular, the question in [1, Final Remarks, item 5] has an affirmative answer.*

Proof. Let $\mu \in M(\mathbb{D})$ satisfy $\widehat{\mu}(\Gamma) \subset K$, and choose an integer $N \geq \max\{1, \|\mu\|\}$. Put

$$A_j := \{\gamma \in \Gamma : \widehat{\mu}(\gamma) = a_j\}.$$

Because $K \subset \mathbb{R}$, the transform of μ is real.

The finite prefix

The polynomial estimate gives $\|q_N(\mu)\| \le B_N$. By (3), $q_N(\widehat{\mu})$ is within $\delta(1,1,B_N)$ of the integers: it equals j on A_j for $j \le N$, and has absolute value at most that threshold on every later level. Lemma 1 therefore gives $\eta_N \in M(\mathbb{D})$ with

$$\widehat{\eta}_N = \sum_{j=1}^{N} j\, 1_{A_j}, \qquad \|\eta_N\| \le 2B_N + 1 = E_N.$$

Set

$$e_j := p_{j,N}(\eta_N) \qquad (1 \le j \le N).$$

Then $\widehat{e_j} = 1_{A_j}$ and $\|e_j\| \le C_N$. In particular, the e_j are idempotent and pairwise convolution-orthogonal. The residual

$$\tau_{N+1} := \mu - \sum_{j=1}^{N} a_j e_j$$

satisfies

$$\|\tau_{N+1}\| \le \|\mu\| + C_N \sum_{j=1}^{N} a_j \le T_N \le a_{N+1}^{-1/2},$$

and its transform is $a_j 1_{A_j}$ on levels $j > N$ and zero elsewhere.

The infinite tail

Inductively suppose $n \ge N+1$, the idempotents e_j have been defined for $j < n$, and

$$\tau_n := \mu - \sum_{j<n} a_j e_j, \qquad \|\tau_n\| \le a_n^{-1/2}.$$

The transform of $a_n^{-1}\tau_n$ is 1 on A_n, is zero on the earlier levels, and has absolute value at most a_{n+1}/a_n on all later levels. Its norm is at most $a_n^{-3/2}$. By (2) and Lemma 1 with $\varepsilon_1 = \varepsilon_2 = 1/2$, there is an idempotent e_n with

$$\widehat{e}_n = 1_{A_n}, \qquad \|e_n\| \le \tfrac{3}{2} a_n^{-3/2} + \tfrac{1}{2}.$$

Define $\tau_{n+1} := \tau_n - a_n e_n$. Since $a_n \le 1$,

$$\|\tau_{n+1}\| \le a_n^{-1/2} + a_n\left(\tfrac{3}{2}a_n^{-3/2} + \tfrac{1}{2}\right) \le 3a_n^{-1/2} \le a_{n+1}^{-1/2},$$

where the last inequality follows from (1). This completes the induction. Fourier uniqueness also shows that all e_j are pairwise orthogonal, including across the prefix/tail boundary.

Norm-convergent powers

For $m = 2, 3$, the series $\sum_j a_j^m e_j$ converges in measure norm. Indeed, the prefix is finite and, on the tail,

$$a_j^m \|e_j\| \le \tfrac{3}{2} a_j^{m-3/2} + \tfrac{1}{2} a_j^m,$$

whose sum converges because $a_{j+1} < a_j/9$. Its Fourier transform is $\widehat{\mu}^m$, so Fourier uniqueness yields

$$\mu^{*m} = \sum_{j \ge 1} a_j^m e_j, \qquad m = 2, 3. \tag{5}$$

Let φ be a character of the commutative Banach algebra $M(\mathbb{D})$ and put $z := \varphi(\mu)$. Each $\varphi(e_j)$ is 0 or 1, and orthogonality permits at most one index with value 1. If all are zero, (5) gives $z^2 = 0$, hence $z = 0$. Otherwise, for the unique index j with $\varphi(e_j) = 1$, it gives

$$z^2 = a_j^2, \qquad z^3 = a_j^3,$$

and therefore $z = a_j$. Moreover, $\varphi(e_j) = 1$ implies $e_j \neq 0$; as $\widehat{e}_j = 1_{A_j}$, Fourier uniqueness then implies $A_j \neq \varnothing$. Thus every nonzero spectral value is an actual Fourier value.

It remains only to handle a possible spectral value 0. If 0 lies in $\overline{\widehat{\mu}(\Gamma)}$, there is nothing to prove. Otherwise only finitely many levels A_j are nonempty and 0 is not itself a Fourier value. If J is that finite set of indices, the polynomial $P(t) = \prod_{j \in J}(t - a_j)$ satisfies $\widehat{P(\mu)} = 0$, hence $P(\mu) = 0$. Spectral mapping then excludes 0 from $\sigma(\mu)$ because $P(0) \neq 0$. Consequently

$$\sigma(\mu) \subset \overline{\widehat{\mu}(\Gamma)}.$$

The reverse inclusion always holds because Fourier evaluations are characters and the spectrum is closed. Hence equality holds, completing the proof. □

6 Verification and novelty status

The proof is qualitative: it uses only positivity of the rounding thresholds, so no numerical estimate of a_n is required. The recursive order is non-circular: q_N, B_N, E_N, C_N, T_N depend only on $a_1, \ldots, a_N$, and all four conditions on a_{N+1} can therefore be imposed simultaneously.

The two principal audit points are:

1. the lifted finite-quotient measures have exactly zero Fourier transform outside Γ_r, and weak-star convergence fixes every individual coefficient eventually;
2. the cube in (5) is essential: the square alone would allow the extraneous value $-a_j$.

A bounded novelty search was performed through 11 August 2026 in the run indexes, arXiv, the journal/publisher record, and exact web searches using "Cantor group", "infinite Wiener–Pitt set", arXiv:1305.3324, the source authors, and arXiv:2510.24578. The only direct continuation found is [2], which assumes both strong continuity and $\|\mu\| \leq 1$ and explicitly says its method does not remove strong continuity. No all-measures answer to the Cantor-group question was found. The novelty claim remains provisional pending expert literature review.

7 Human-review recommendation

Prioritize review of Lemma 1, especially the identification of the finite-group A-norm with the total variation norm of the pushed-forward measure, and the simultaneous prefix isolation at stage N. If those steps are accepted, the remaining induction and Gelfand-spectrum argument are elementary and close the question as stated.

References

[1] P. Ohrysko and M. Wojciechowski, *On the relationships between Fourier–Stieltjes coefficients and spectra of measures*, Studia Math. 221 (2014), 117–140; arXiv:1305.3324.

[2] P. Ohrysko, T. Sanders, and M. Wojciechowski, *Wiener–Pitt sets for compact Abelian groups*, J. Fourier Anal. Appl. 32 (2026), Article 30; arXiv:2510.24578; DOI 10.1007/s00041-026-10239-1.